

Technical datasheet

GENESTAR™ GP2450NH-2

GENESTAR™ is a brand name of heat resistant polyamide developed by Kuraray Co., Ltd. GP2450NH-2 is a standard reinforced injection molding grade for E&E applications provided as pellets. This grade meets UL94, V-O Flammability rating halogen-free (without Chlorine and Bromine)*.

*Halogen free' follows the standards that Br is less than 900ppm, Cl is less than 900ppm, and total of halogen amount (Br and Cl) is less than 1500ppm.

The main features are:

- Mechanical properties
- Flow-ability weld strength

This grade is suitable for

- Board-to-board and wire-to-board connectors
- Insulator for industrial equipment

(Condition: 23°C)

	Value*	Unit	Standard
Glass-fiber content	45	%	
Physical properties			
Specific gravity	1.53	g cm ⁻³	ISO 1183
Water absorption (23°C, 24h)	0.10	%	ISO 62
Mechanical properties			
Tensile strength	146	MPa	ISO 527
Tensile elongation at break	1.5	%	ISO 527
Tensile modulus	15.1	GPa	ISO 527
Flexural strength	220	MPa	ISO 178
Flexural modulus	13.9	GPa	ISO 178
Weld line strength	30	MPa	Kuraray method based on ISO527
Charpy impact strength - notched	9.3	kJ m ⁻²	ISO 179/1eA
Thermal properties			
DTUL (1.8 MPa)	280	°C	ISO 75Af
Melting point	306	°C	ISO 11357-3
Glass transition temperature	125	°C	ISO 11357-2
Molding properties			
Mold shrinkage – MD	0.2	%	ISO 294-4
Mold shrinkage – TD	0.7	%	ISO 294-4
Bar-flow length at 0.5 mm	55	mm	750 kgf, 320°C
Flammability			
Flame rating	V-O	(0.15, 0.4-0.44mm)	UL94
GWFI (t = 0.75, 1.5, 3.0 mm)	960, 930, 960	°C	IEC60695-2 10~13
GWIT (t = 0.75, 1.5, 3.0 mm)	750, 750, 825	°C	IEC60695-2 10~13

	Value*	Unit	Standard
Electrical properties			
Dielectric strength	33	kV/mm	IEC 60243-1
Tracking resistance (CTI)	0	PLC-class	UL746A
Inclined Plane Tracking (IPT)	1.5	kV	UL746A
HWI (0.4, 1.5, 3.0mm)	3, 0, 0	PLC	UL746A
HAI (0.4, 1.5, 3.0mm)	3, 3, 0	PLC	UL746A
Dielectric loss tangent (1 MHz)	0.0134	-	JIS C2138
(1 GHz)	0.0086	-	ASTM D2520
(10 GHz)	0.0088	-	ASTM D2520

* all values are measured in dry condition as molded

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